(KTXFI2EBNF) Physics II. Homework No. 4

December 18, 2025

1. By how much does the emitted current change if the tungsten cathode temperature is increased from $T = 2250\,\mathrm{K}$ to $2400\,\mathrm{K}$? The work function for tungsten is $W_{\mathrm{ki}} = 5.0\,\mathrm{eV}$. The Richardson coefficient is $6.05 \cdot 10^5\,\mathrm{A/m^2K^2}$.
2. What retarding voltage reduces the emission current density to one fifteenth for a cathode at $T = 2300\,\mathrm{K}$? The electron charge is $q = -1.6 \cdot 10^{-19}\,\mathrm{As}$.
3. What is the free-electron drift velocity in a copper conductor of radius $R = 2.5\,\mathrm{mm}$ if the current is $I = 15\,\mathrm{A}$? The electron density in copper is $n_{\mathrm{e}} = 3.3 \cdot 10^{28}\,\mathrm{m^{-3}}$.
4. We examine an n-type semiconductor under the conditions shown in the figure: $B_y = 0.5\,\mathrm{T}$, $U_{\mathrm{H}} = 12\,\mathrm{mV}$, $I = 25\,\mathrm{mA}$. The magnetic induction is directed along the y–axis.
   (a) What is the electron concentration?
   (b) What is the drift velocity of the electrons?
   (c) What is the Hall constant?

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